

Chapter 1 Descriptive statistics

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- 2 Basic concepts
- 3 Sampling schemes
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Introduction about the course

- Books

- Ramachandran K M, Tsokos C P. Mathematical statistics with applications in R (3rd edition). Academic Press, 2021.

- References

- Wackerly D, Mendenhall W, Scheaffer R L. Mathematical statistics with applications. Cengage Learning, 2014.

- Homework

- Please use $\text{\LaTeX}$ to write all the homework in this course;
- Any programming language is ok, R language is welcome;
- Please submit the homework on time according to the requirement;
 - PDF document;
 - code script;
 - $\text{\LaTeX}$ source file folder (.rar).

Objective

Three fundamental problems in statistics (R A Fisher):

- Specification of the type of population that the data came from
 - The choice of the mathematical form of the population
- Estimation
 - Calculating the statistics to estimate the unknown parameters
- Distribution
 - Mathematical deduction of the exact nature of the distributions
 - Test the validity of our specification (tests of Goodness of Fit)

Earlier work on statistics

- Collection;
- Organization;
- Presentation of data in the form of tables and charts.

Modern applications of statistical techniques

Statistical methods can aid us in several ways:

- Design experiments and surveys;
- Descriptive statistics: organizing, describing, summarizing, and displaying experimental data;
 - More general area: exploratory data analysis.
- Draw inferences and make decisions based on data.

How to solve a real-life problem using statistics:

- Identify the model using the appropriate statistical method;
- Justify the applicability of the selected mode;
- Apply the related model to analyze the data and make the necessary decisions.

One of the first problems that a statistician faces is obtaining the data.

Steps for data collection:

- Define the objectives of the problem and proceed to develop the experiment or survey;
- Define the variables or parameters of interest;
- Define the procedures of data-collection and -measuring techniques.
 - Sampling procedures
 - Sample size
 - Data-measuring devices (questionnaires, telephone interviews, etc.)

Example 1.1.1 How to collect data to estimate the average household income in a certain community?

- To acquire the data, we may send out a questionnaire or conduct a telephone interview.
- Once we have the data, we may first want to represent the data in graphical or tabular form to better understand its distributional behavior.
- Then we will use appropriate analytical techniques to estimate the parameter(s) of interest, in this case the average household income.

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Some basic notions

- Definition 1.2.1 Population

A population is the collection or set of all objects or measurements that are of interest to the collector.

- Definition 1.2.2 Sample

The sample is a subset of data selected from a population.

- Definition 1.2.2 Sample size

The size of a sample is the number of elements in it.

- Definition 1.2.3 Statistical inference

A statistical inference is an estimate, a prediction, a decision, or a generalization about the population based on information contained in a sample

Some examples

- Political polls
- Laboratory experiment
- Clinical studies
- Finance

Types of data

- Definition 1.2.4 Quantitative data

Observations measured on a numerical scale

- Definition 1.2.4 Qualitative or categorical data

Nonnumerical data that can only be classified into one of the groups of categories

- nominal data - do not have a specific order
- ordinal data - listed in a specific order

- Definition 1.2.5 Cross-sectional data

Data collected on different elements or variables at the same point in time or for the same period of time

- Definition 1.2.6 Time series data

Data collected on the same element or the same variable at different points in time or for different periods of time

Types of data

Example 1.2.3:

- The number of minority-owned businesses in Florida is quantitative data.
- The number of car accidents in different U.S. cities is quantitative data.
- The blood group of each person in a community as O, A, B, AB is qualitative data.
- The marital status of each person in a statistics class as married or not married is qualitative or categorical data.

Types of data

Example 1.2.4: The data in Table 1.1 represent U.S. federal support for the mathematical sciences in 1996, in millions of dollars (source: AMS Notices). This is an example of cross-sectional data, as the data are collected in one time period, namely in 1996.

TABLE 1.1 Federal Support for the Mathematical Sciences, 1996.	
Federal agency	Amount
National Science Foundation	91.70
DMS	85.29
Other MPS	4.00
Department of Defense	77.30
AFOSR	16.70
ARO	15.00
DARPA	22.90
NSA	2.50
ONR	20.20
Department of Energy	16.00
University Support	5.50
National Laboratories	10.50
Total, all agencies	185.00

Types of data

Example 1.2.5: The data in Table 1.2 represent U.S. federal support for the mathematical sciences during the years 1995-97, in millions of dollars (source: AMS Notices). This is an example of time series data, because they have been collected at different time periods, 1995 through 1997.

TABLE 1.2 United States Federal Support for the Mathematical Sciences in Different Years.			
Agency	1995	1996	1997
National Science Foundation	87.69	91.70	98.22
DMS	85.29	87.70	93.22
Other MPS	2.40	4.00	5.00
Department of Defense	77.40	77.30	67.80
AFOSR	17.40	16.70	17.10
ARO	15.00	15.00	13.00
DARPA	21.00	22.90	19.50
NSA	2.50	2.50	2.10
ONR	21.40	20.20	16.10
Department of Energy	15.70	16.00	16.00
University Support	6.20	5.50	5.00
National Laboratories	9.50	10.50	11.00
Total, all agencies	180.79	185.00	182.02

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Representative Samples

- It is usually not possible to obtain information on the entire population because the population is too large to attempt a survey of all of its members, or it may not be cost-effective;
- A small but carefully chosen sample can be used to represent the population.
- **Representative sample**
A sample that accurately reflects its population characteristics
- **Biased sample**
A sample that is not representative of the population characteristics

The reliability or accuracy of conclusions drawn concerning a population depends on whether or not the sample is properly chosen so as to represent the population sufficiently well. **The choice between these sampling methods depends on:**

- the nature of the problem or investigation
- the availability of good sampling frames (a list of all of the population members)
- the budget or available financial resources
- the desired level of accuracy
- the method by which data will be collected, such as questionnaires or interviews.

Simple random sample

- **Definition 1.3.1** A **simple random sample** is one selected in such a way that every element of the population has an equal chance of being chosen;
- Equivalently, each possible sample of size n has the same chance of being selected as any other subset of sample of size n ;

Some advantages of simple random sampling:

- Selection of sampling observations at random ensures against possible investigator biases;
- Analytic computations are relatively simple, and probabilistic bounds on errors can be computed in many cases;
- It is frequently possible to estimate the sample size for a prescribed error level when designing the sampling procedure.

- **EXAMPLE 1.3.1**

- For a state lottery, 52 identical ping-pong balls with a number from 1 to 52 painted on each ball are put in a clear plastic bin.
- A machine thoroughly mixes the balls and then six are selected.
- The six numbers on the chosen balls are the six lottery numbers that have been selected by a simple random sampling procedure.

Systematic sampling

- **Definition 1.3.2** A **systematic sample** is a sample in which every K_{th} element in the sampling frame is selected after a suitable random start for the first element.
- We list the population elements in some order (say alphabetical) and choose the desired sampling fraction.

Steps for selecting a systematic sample:

- Number the elements of the population from 1 to N .
- Decide on the sample size, say n , that we need.
- Choose $K = N/n$.
- Randomly select an integer between 1 and K .
- Then take every K_{th} element.

- Example 1.3.2

If the population has 1000 elements arranged in some order and we decide to sample 10% (i.e., $N = 1000$ and $n = 100$), then $K = \frac{1000}{100}$. Pick a number at random between 1 and $K = 10$ inclusive, say 3. Then select elements numbered 3, 13, 23, $\dots$, 993.

- Remark of Systematic sampling

Systematic sampling is widely used because it is easy to implement. If the population elements are ordered, systematic sampling is a better sampling method.

- Definition 1.3.3

A **stratified sample** is one obtained by stratifying (dividing into nonoverlapping groups) the sampling frame based on some factors and then selecting some elements from each of the strata;

- A population with N elements is divided into s subpopulations, and a sample is drawn from each subpopulation independently.
- The size and sample size of each subpopulation may vary.

Stratified sampling

- Steps for selecting a stratified sample:
 - Decide on the relevant stratification factors (sex, age, income, etc.).
 - Divide the entire population into strata (subpopulations) based on the stratification criteria. Sizes of strata may vary.
 - Select the requisite number of units using simple random sampling or systematic sampling from each subpopulation.
- **Example 1.3.3** In a population of 1000 children from an area school, there are 600 boys and 400 girls. We divide them into strata based on their parents' income as shown in Table 1.3.

TABLE 1.3 Classification of School Children.

	Boys	Girls
Poor	120	240
Middle class	150	100
Rich	330	60

This is stratified data.

Stratified sampling

- **Example 1.3.4** Refer to Example 1.3.3. Suppose we decide to sample 100 children from the population of 1000 (that is, 10% of the population). We also choose to sample 10% from each of the categories. For example, we would choose 12 (10% of 120) poor boys; 6 (10% of 60 rich girls) and so forth. This yields Table 1.4. This particular sampling method is called a **proportional stratified sampling**.

TABLE 1.4 Proportional Stratification of School Children.

	Boys	Girls
Poor	12	24
Middle class	15	10
Rich	33	6

- Some uses of stratified sampling
 - In addition to providing information about the whole population, this sampling scheme provides information about the subpopulations, the study of which may be of interest.
 - Stratified sampling can be considerably more precise than a simple random sample, because the population is fairly homogeneous within each stratum but there is a sizable variation between the strata.

- Definition 1.3.4 Cluster sampling (also called area sampling)
 - The sampling unit contains groups of elements called clusters instead of individual elements of the population.
 - Unlike the stratified sample where the strata are created by the researcher based on stratification variables, the clusters naturally exist and are not formed by the researcher for data collection.
- EXAMPLE 1.3.5
 - Suppose we wish to select a sample of about 10% from all fifth-grade children of a county. We randomly select 10% of the elementary schools assumed to have approximately the same number of fifth-grade students and select all fifth-grade children from these schools.
 - This is an example of cluster sampling, each cluster being an elementary school that was selected.

- Definition 1.3.5 Multiphase sampling

- It involves collection of some information from the whole sample and additional information either at the same time or later from subsamples of the whole sample.
- The multiphase or multistage sampling is basically a combination of the techniques presented earlier.

- EXAMPLE 1.3.6

- An investigator in a population census may ask basic questions such as sex, age, or marital status for the whole population, but only 10% of the population may be asked about their level of education or about how many years of mathematics and science education they had.

The sample observations are prone to various sources of error that may seriously affect the inferences about the population

- **Sampling errors**

errors due to the fact that the sample is not an exact representative of the population

- **Nonsampling errors**

errors occur in the collection, recording and processing of sample data

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Visual displays of data

- The source of our statistical knowledge lies in the data;
- Our interest in the data lies with the story it tells;
- These visual displays may reveal the patterns of behavior of the variables being studied.
- Most common graphical displays
 - frequency table
 - pie chart
 - bar graph
 - Pareto chart
 - histogram
 - ...

Bar graph

- **Definition 1.4.1** It is a graph of bars whose heights represent the frequencies (or relative frequencies) of respective categories.
- **Example 1.4.1:** The data in the following table represent the percentages of price increases of some consumer goods and services for the period December 1990 to December 2000 in a certain city. Construct a bar chart for these data.

TABLE 1.5 Percentages of Price Increases of Some Consumer Goods and Services.

Medical care	83.3%
Electricity	22.1%
Residential rent	43.5%
Food	41.1%
Consumer price index	35.8%
Apparel and upkeep	21.2%

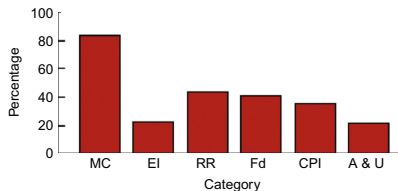


FIGURE 1.1 Percentage price increase of consumer goods.

Pareto chart

- Vilfredo Pareto (1848 - 1923): Pareto effect or 80/20 rule
- Reflect the relative importance of different factors;
- The height of the bars proportional to the contribution of each factor;
- The bars are displayed from the most numerous category to the least.

Example 1.4.2. For the data of Example 1.4.1, construct a Pareto chart.

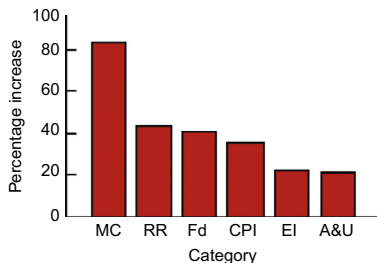


FIGURE 1.2 Pareto chart.

Pie graph

- **Definition 1.4.2** It is a circle divided into sectors that represent the percentages of a population or a sample that belongs to different categories;
 - Pie charts are especially useful for presenting categorical data;
 - The pie “slices” are drawn such that they have an area proportional to the frequency.
- **Example 1.4.3** The combined percentages of carbon monoxide (CO) and ozone (O₃) emissions from different sources are listed in Table 1.6. Construct a pie chart.

TABLE 1.6 Combined Percentages of CO and O₃ Emissions.

Transportation (T)	Industrial process (I)	Fuel combustion (F)	Solid waste (S)	Miscellaneous (M)
63%	10%	14%	5%	8%

Pie graph

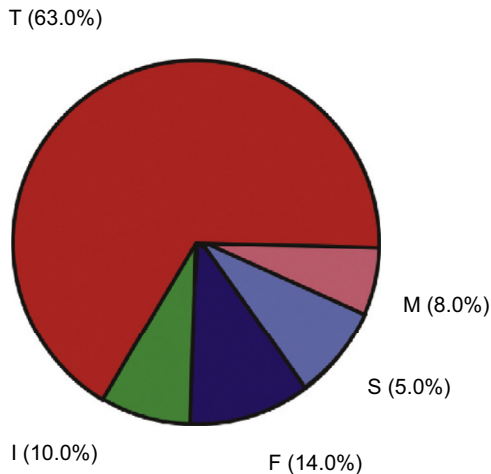


FIGURE 1.3 Pie chart for CO and O₃.

Stem-and-leaf plot

- **Definition 1.4.3** A **stem-and-leaf plot** is a simple way of summarizing quantitative data;
- When data sets are relatively small, it is particularly useful.
- Each data value is split into a “stem” and a “leaf.”
 - The “leaf” is usually the last digit of the number;
 - The other digits to the left of the “leaf” form the “stem”.
- **Example 1.4.4** Construct a stem-and-leaf plot for the 20 test scores given below.

78	74	82	66	94	71	64	88	55	80
91	74	82	75	96	78	84	79	71	83

Stem-and-leaf plot

- Use the first digit of the score as the stem;
- Use the second digit as the leaf.

TABLE 1.7 Stem-and-Leaf Display of 20 Exam Scores.

Stem	Leaves							
5	5							
6	6	4						
7	8	4	1	4	5	8	9	1
8	2	8	0	2	4	3		
9	4	1	6					

- We can identify the shape and distribution of data such as the symmetry;
- We can see where the maximum and minimum are located with respect to the frequencies;
- We can know whether they are bell shaped, which is of paramount

Frequency table

- A **frequency table** is a table that divides a data set into a suitable number of categories (classes);
- Data presented in the form of a frequency table: grouped data;
- Generally, the classes will be intervals of equal length.
- Let f_i denote the frequency of the class i and let n be sum of all frequencies.
 - The **relative frequency** for the class i :

$$\frac{f_i}{n}$$

- The **cumulative relative frequency** for the class i :

$$\sum_{k=1}^i \frac{f_k}{n}$$

Frequency table

- **Example 1.4.5** The following data give the lifetime of 30 incandescent light bulbs (rounded to the nearest hour) of a particular type. Construct a frequency, relative frequency, and cumulative relative frequency table.

872	931	1146	1079	915	879	863	1112	979	1120
1150	987	958	1149	1057	1082	1053	1048	1118	1088
868	996	1102	1130	1002	990	1052	1116	1119	1028

Class	Frequency f_i	Relative frequency $\frac{f_i}{\sum f_i}$	Cumulative relative frequency $\sum_{k=1}^i \frac{f_k}{n}$
50–900	4	4/30	4/30
900–950	2	2/30	6/30
950–1000	5	5/30	11/30
1000–1050	3	3/30	14/30
1050–1100	6	6/30	20/30
1100–1150	10	10/30	30/30

Histogram

- Classes are marked on the horizontal axis;
- Either the frequencies, relative frequencies, or percentages are represented by the heights on the vertical axis;
- The bars are drawn adjacent to each other without any gaps.
- Histograms are great to identify range and skew of **quantitative** data;
- It is useful when the number of observations is relatively large, and there are no natural separate categories or classes;
- It shows the location of the mean and modes of the data and the variation in the data, especially the range.

Steps for the construction of a frequency table and histogram

- Determine the range of the observations

$$R = \text{maximum value} - \text{minimum value}.$$

- Determine the number of classes and the class width
 - class number: from 5 to 20
 - The class width should be slightly **larger** than the ratio

$$\frac{\text{Largest value} - \text{Smallest value}}{\text{Number of classes}}$$

- Construct a table (frequency table) that lists
 - The class intervals
 - A tabulation of the number of measurements in each class (tally)
 - The frequency f_i of each class
 - Relative frequency f_i/n . (if needed)
- Draw bars over each interval with heights being the frequencies (or relative frequencies).

Some points:

- The first interval should begin a little below the minimum value, and the last interval should end a little above the maximum value.
- The intervals are called class intervals and the boundaries are called class boundaries.
- The class limits are the smallest and the largest data values in the class.
- The class mark is the midpoint of a class.
- None of the data values should fall on the boundaries of the classes.

Histogram

Example 1.4.6 The following data refer to a certain type of chemical impurity measured in parts per million in 25 drinking-water samples randomly collected from different areas of a county.

- Make a frequency table displaying class intervals, frequencies, relative frequencies, and percentages.
- Construct a frequency histogram.

11 19 24 30 12 20 25 29 15 21 24 31

16 23 25 26 32 17 22 26 35 18 24 18 27

Solution We will use five classes. The maximum and minimum values in the data set are 35 and 11. the class width is $\frac{(35-11)}{5} \simeq 5$. we shall take the class width to be 5. The lower boundary of the first class interval will be chosen to be 10.5. With five classes, each of width 5, the upper boundary of the fifth class becomes 35.5.

Frequency table and Histogram

- **Solution of Example 1.4.6** We can now construct the frequency table and generate a histogram for the data.

Class	Class interval	f_i = frequency	Relative frequency	Percentage
1	10.5–15.5	3	$3/25 = 0.12$	12
2	15.5–20.5	6	$6/25 = 0.24$	24
3	20.5–25.5	8	$8/25 = 0.32$	32
4	25.5–30.5	5	$5/25 = 0.20$	20
5	30.5–35.5	3	$3/25 = 0.12$	12

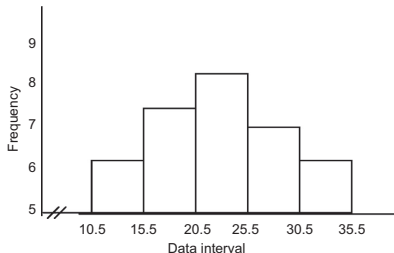


FIGURE 1.4 Frequency histogram of impurity data.

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Suppose that we have a sample with values $x_1, x_2, \dots, x_n$. Some characteristics associated with this data set:

- Central tendency
 - sample mean
 - median
 - mode
- Variability
 - sample variance
 - sample standard deviation
 - interquartile range

Definition 1.5.1 Let $x_1, x_2, \dots, x_n$ be a set of sample values.

- **Sample mean** (or empirical mean)

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

- **Sample variance**

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- **Sample standard deviation**

$$s = \sqrt{s^2}$$

Some points

- The sample variance and the sample standard deviation both are measures of the variability or “scatteredness” of data values around the sample mean.
- The larger the variance, the greater is the spread.
- **Theorem 1.5.1** For a given set of measurements $x_1, x_2, \dots, x_n$. Let $\bar{x}$ be the sample mean. Then

$$\sum_{i=1}^n (x_i - \bar{x}) = 0$$

- A simple computational formula of s^2

$$s^2 = \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - \frac{1}{n} \left(\sum_{i=1}^n x_i \right)^2 \right] = \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right]$$

Other central tendency measures

- Definition 1.5.2 Median (Q_2)

- Odd number of elements: the middle number of the ordered set
- Even number of elements: the average of the middle two numbers

- Trimmed mean

For $0 < \alpha < 1$, a $100\alpha\%$ trimmed mean is found as follows:

- Order the data
- Discard the lowest $100\alpha\%$ and the highest $100\alpha\%$ of the data values.
- Find the mean of the rest of the data values (denoted by $\bar{x}_\alpha$).

- Definition 1.5.3 Mode:

- The most frequently occurring member of the data set.
- If all the data values are different, the data set has no mode.

Other measures of dispersion

- **Average deviation** one of other measures of dispersion

$$\text{Average deviation} = \frac{\sum_{i=1}^n |x_i - \bar{x}|}{n}$$

- **Definition 1.5.2 Interquartile range**

- The **lower quartile** (Q_1): the middle number of the half of the data below the median
- The **upper quartile** (Q_3): the middle number of the half of the data above the median
- The **interquartile range** (IQR):

$$IQR = Q_3 - Q_1$$

- A possible **outlier (mild outlier)** detection method: Any data point that lies below $Q_1 - 1.5(IQR)$ or above $Q_3 + 1.5(IQR)$

Examples

Example 1.5.1 The following data give the time in months from hire to promotion to manager for a random sample of 25 software engineers from all software engineers employed by a large telecommunications firm. Calculate the mean, median, mode, variance, and standard deviation for this sample.

5 7 229 453 12 14 18 14 14 483
22 21 25 23 24 34 37 34 49 64
47 67 69 192 125

Example 1.5.2 For the data of Example 1.5.1, find lower and upper quartiles, median, and interquartile range (IQR). Check for any outliers.

Example 1.5.3 For the data set representing the number of children in a random sample of 10 families in a neighborhood, find the 10% trimmed mean ($\alpha = 0.1$).

1 2 2 3 2 3 9 1 6 2

Numerical measures for grouped data

Let the grouped data have l classes, with m_i being the midpoint and f_i being the frequency of class $i, i = 1, 2, \dots, l$. Let $n = \sum_{i=1}^l f_i$.

Definition 1.5.4

- The approximate values for $\bar{x}$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^l f_i m_i$$

Numerical measures for grouped data

- The approximate values for s^2

$$\begin{aligned}s^2 &= \frac{1}{n-1} \sum_{i=1}^l f_i (m_i - \bar{x})^2 \\&= \frac{1}{n-1} \left(\sum_{i=1}^l f_i m_i^2 - \sum_{i=1}^l 2f_i m_i \bar{x} + \sum_{i=1}^l f_i \bar{x}^2 \right) \\&= \frac{1}{n-1} \left(\sum_{i=1}^l f_i m_i^2 - n\bar{x}^2 \right) \\&= \frac{1}{n-1} \sum_{i=1}^l \left[f_i m_i^2 - \frac{1}{n} \left(\sum_{i=1}^l f_i m_i \right)^2 \right]\end{aligned}$$

Numerical measures for grouped data

- The approximate values of median
 - L = lower class limit of the interval that contains the median.
 - n = total frequency.
 - F_b = cumulative frequencies for all classes before the median class.
 - f_m = frequency of the class interval containing the median.
 - w = interval width of the interval that contains the median.
 - The median for the grouped data is given by

$$M = L + \frac{w}{f_m}(0.5n - F_b)$$

Example 1.5.4 The grouped data in the following table represent the number of children from birth through the end of the teenage years in a large apartment complex. Find the mean, variance, and standard deviation for these data.

Class	0-3	4-7	8-11	12-15	16-19
Frequency	7	4	19	12	8

Examples

Solution

Note that even though the classes are given as disjoint, in actuality these are adjacent age intervals, like $[0, 4)$, $[4, 8)$, etc. When we take the class midpoint, we have to take this into account. For simplicity of calculation we create Table 1.19.

The sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^l f_i m_i = \frac{540}{50} = 10.80.$$

The sample variance is

$$s^2 = \frac{\sum_{i=1}^l f_i m_i^2 - \frac{(\sum_{i=1}^l f_i m_i)^2}{n}}{n - 1} = \frac{7016 - \frac{(540)^2}{50}}{49} = 24.1632650 \approx 24.16.$$

The sample standard deviation is $s = \sqrt{s^2} = 4.9156144 \approx 4.92$.

TABLE 1.19 Summary Statistics for Number of Children.

Class	<i>Interval</i>	f_i	m_i	$m_i f_i$	$m_i^2 f_i$
0–3	[0, 4)	7	2	14	28
4–7	[4, 8)	4	6	24	144
8–11	[8, 12)	19	10	190	1900
12–15	[12, 16)	12	14	168	2352
16–19	[16, 20)	8	18	144	2592
	$n = 50$		$\sum m_i f_i = 540$	$\sum m_i^2 f_i = 7016$	

Examples

Example 1.5.5 For the data in Example 1.5.4, find the median.

Solution

First, we develop Table 1.20.

TABLE 1.20 Frequency Distribution for Number of Children.

Class	f_i	Cumulative f_i	Cumulative f_i/n
0–3	7	7	0.14
4–7	4	11	0.22
8–11	19	30	0.6
12–15	12	42	0.84
16–19	8	50	1.00

Examples

The first interval for which the cumulative relative frequency exceeds 0.5 is the interval that contains the median. Hence, the interval 8 to 11 contains the median. Therefore, $L = 8$, $f_m = 19$, $n = 50$, $w = 3$, and $F_b = 11$. Then, the median is

$$M = L + \frac{w}{f_m}(0.5n - F_b) = 8 + \frac{3}{19}((0.5)(50) - 11) = 10.211.$$

It is important to note that all the numerical measures we calculate for grouped data are only approximations to the actual values of the ungrouped data if they are available.

the use of the sample standard deviation

- Empirical rule

When the histogram of a data set is "bell-shaped" or "moundshaped," and symmetric, the empirical rule states:

- Approximately 68% of the data are in the interval $\bar{x} - s, \bar{x} + s$
- Approximately 95% of the data are in the interval $\bar{x} - 2s, \bar{x} + 2s$
- Approximately 99.7% of the data are in the interval $\bar{x} - 3s, \bar{x} + 3s$

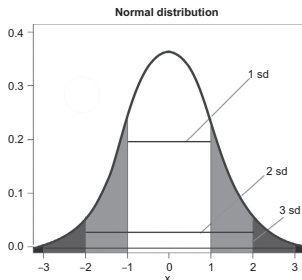


FIGURE 1.5 Bell-shaped curve.

- The sample mean or the sample standard deviation focuses on a single aspect of the data set.
- The histograms and stem-and-leaf displays express rather general ideas about the data.
- A box plot (also called box-and-whisker plots) can be used to describe several prominent features of a data set
 - minimum
 - lower quartile
 - median
 - upper quartile
 - maximum

Procedure to construct a box plot

- Draw a vertical measurement axis and mark Q_1 , Q_2 (median), and Q_3 on this axis. Let $IQR = Q_3 - Q_1$.
- Construct a rectangular box whose bottom edge lies at the lower quartile, Q_1 , and whose upper edge lies at the upper quartile, Q_3 .
- Draw a horizontal line segment inside the box through the median.
- Extend the lines from each end of the box out to the farthest observation that is still within $1.5(IQR)$ of the corresponding edge: **whiskers**.
- Draw an open circle (or asterisks *) to identify each observation that falls between $1.5(IQR)$ and $3(IQR)$ from the edge to which it is closest: **mild outliers**.
- Draw a solid circle to identify each observation that falls more than $3(IQR)$ from the closest edge: **extreme outliers**.

Box plots

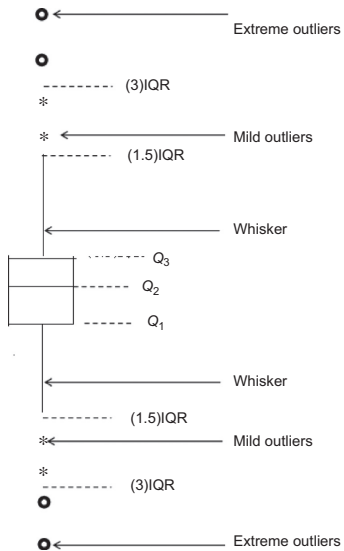


FIGURE 1.6 A typical box-and-whiskers plot.

Example 1.5.6 The following data identify the time in months from hire to promotion to chief pharmacist for a random sample of 25 employees from a certain group of employees in a large corporation of drugstores.

5 7 229 453 12 14 18 14 14 483 22 21
25 23 24 34 37 34 49 64 47 67 69 192 125

- Construct a box plot.
- Do the data appear to be symmetrically distributed along the measurement axis?

Examples

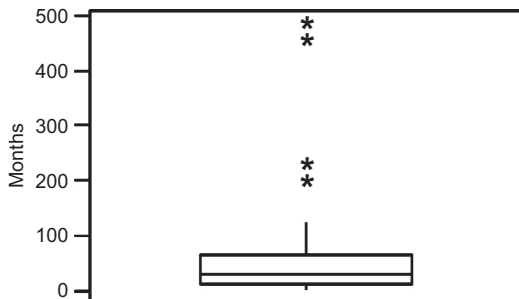


FIGURE 1.7 Box plot for months to promotion.

By examining the relative position of the median line (the middle line in Fig. 1.7), we can test the symmetry of the data. For example, in Fig. 1.7, the median line is closer to the lower quartile than the upper line, which suggests that the distribution is slightly nonsymmetrical. Also, a look at this box plot shows the presence of two mild outliers and two extreme outliers.

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- For small sets of data, many basic calculations such as finding means and standard deviations and creating simple charts, graphing calculators are sufficient.
- For deeper analysis and for large data sets, statistical software is necessary.
 - R
 - MatLab
 - SAS
 - Minitab
 - SPSS
 - SAS

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- Some basic definitions: population and sample
- Some sampling techniques
- Some graphical presentations
- Some descriptive statistics: mean, median, and variance
- A brief discussion of the technology and statistics

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R introduction and examples

- R is a free software for statistical computing and graphics
- Website: <http://www.r-project.org/>
- IDE: R-studio (<https://www.rstudio.com/>)
- Some references for R
 - <http://www.ecostat.unical.it/tarsitano/Didattica/LabStat2/Everitt.pdf>

Projects for chapter 1

- 1A World Wide Web and data collection
 - Find an open real data set, and pick any category of interest to you and obtain data.
 - Represent a section of the data graphically.
 - Find mean, median, and standard deviation.
 - Identify any outliers.
- 1B Preparing a list of useful Internet sites

Prepare a list of Internet references for various aspects of statistical study.

Projects for chapter 1

- 1C Dot plots and descriptive statistics

From the local advertisements of apartments for rent, randomly pick 50 monthly rents for two-bedroom apartments. For these data, first draw a dot plot and then obtain descriptive statistics (use R, or any other statistical software).

- 1D Importance of statistics in our society

Write a short report on the importance of statistics in our modern-day society. Give different examples to illustrate your points. One interesting project will be to study the role of the Internet of Things (IoT), a vast network of smart objects that work together in collecting and analyzing data and autonomously performing actions.

- 1E Uses and misuses of statistics

“There are three types of lies: lies, damn lies, and statistics” - Benjamin Disraeli. Write a short report on uses and misuses of statistics.

- 1.4.3, 1.4.14, 1.5.5, 1.5.7, 1.5.10